Supplemental Materials Details

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I have uploaded all supplemental materials in a file named Supplemental Materials Details.zip. Upon unzipping this file, you will find 3 different folders:

1. 1_Modified VLAT
2. 2_Original VLAT
3. 3_Kim_VisQA

I will describe the contents of each folder, including file types, required software for viewing/execution, and how the material relates to the paper. Below is a detailed description of each folder and its contents:

1 1_Modified VLAT

This folder contains two subfolders that correspond to the two different prompting strategies evaluated in our paper:

1.1 1_Generic Prompt

This folder contains materials related to the standard prompting strategy with the modified VLAT charts:

1.1.1 Images

- **Contents:** PNG files of all modified visualization images used in our testing process. These are recreations of the original VLAT charts with randomized data while maintaining the same design style.
- **File type:** PNG
- **Software needed:** Any standard image viewer
- **Relation to the paper:** These images are the modified visualizations described in Section 3.2 of the paper and shown in Figure 1.

1.1.2 Results

This folder contains three subfolders, one for each LLM tested:

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1.1.2.1 1_Claude 3.7_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run_Bad_Prompt_Claude.ipynb`, `New_VLAT_Graphs_Claude_Results_Bad_Prompt.ipynb`
  - CSV files (e.g., `VLAT_1741477361.csv`) containing the model responses, time taken, and correctness (True/False)
  - Corresponding DOCX files (e.g., `VLAT_1741477361.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4.

1.1.2.2 2_GPT 4.5_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run_Bad_Prompt_GPT.ipynb`, `New_VLAT_Graphs_GPT_Results_Bad_Prompt.ipynb`
  - CSV files (e.g., `GPT_VLAT_1741563515.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `GPT_VLAT_1741563515.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4.

1.1.2.3 3_GEMNI 2.0 Pro_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run_Bad_Prompt_Gemni.ipynb`, `New_VLAT_Graphs_Gemni_Results_Bad_Prompt.ipynb`
  - CSV files (e.g., `Gemini_VLAT_1741563773.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `Gemini_VLAT_1741563773.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4.

1.1.3 Additional Files

- `Regenerate VLAT.ipynb`: Jupyter notebook used to generate the modified VLAT visualizations
- `VLAT Questions Metadata.csv`: CSV file containing metadata about the modified VLAT questions
- `VLAT Questions.csv`: CSV file containing the modified VLAT questions and answer options

1.2 2_Charts-of-Thought Prompt

This folder contains materials related to our Charts-of-Thought prompting strategy with the modified VLAT charts:

1.2.1 Images

- **Contents:** Same PNG files as in the Generic Prompt folder, showing all modified visualization images used in testing
- **File type:** PNG
- **Software needed:** Any standard image viewer
- **Relation to the paper:** These images are the modified visualizations described in Section 3.2 of the paper and shown in Figure 1.

1.2.2 Results

This folder contains three subfolders, one for each LLM tested:

1.2.2.1 1_Claude 3.7_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run.ipynb`, `New_VLAT_Graphs_Claude_Results_Good_Prompt.ipynb`
  - CSV files (e.g., `VLAT_1741461195.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `VLAT_1741461195.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4, demonstrating the performance with Charts-of-Thought prompting.

1.2.2.2 2_GPT 4.5_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run_Good_Prompt_GPT.ipynb`, `New_VLAT_Graphs_GPT_Results_Good_Prompt.ipynb`
  - CSV files (e.g., `GPT_VLAT_1741465486.csv`) containing the model responses, time taken, and correctness

- Corresponding DOCX files (e.g., `GPT_VLAT_1741465486.docx`) containing the detailed explanations from the LLM

- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4, demonstrating the performance with Charts-of-Thought prompting.

1.2.2.3 3_GEMNI 2.0 Pro_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run_Good_Prompt_GEMNI.ipynb`, `New_VLAT_Graphs_GEMNI_Results_Good_Prompt.ipynb`
  - CSV files (e.g., `Gemini_VLAT_1741463333.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `Gemini_VLAT_1741463333.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 4, demonstrating the performance with Charts-of-Thought prompting.

1.2.3 Additional Files

- `Regenerate VLAT.ipynb`: Jupyter notebook used to generate the modified VLAT visualizations
- `VLAT Questions Metadata.csv`: CSV file containing metadata about the modified VLAT questions
- `VLAT Questions.csv`: CSV file containing the modified VLAT questions and answer options

2 2_Original VLAT

This folder contains materials related to our experiments with the original VLAT charts:

2.1 Charts-of-Thought Prompt

2.1.1 Images

- **Contents:** PNG files of all original VLAT visualizations
- **File type:** PNG
- **Software needed:** Any standard image viewer
- **Relation to the paper:** These images are the original VLAT visualizations described in Section 5 of the paper.

2.1.2 Results

This folder contains four subfolders, one for each LLM tested:

2.1.2.1 1_Claude 3.7_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run.ipynb`, `Claude_Results.ipynb`
  - CSV files (e.g., `VLAT_1740626876.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `VLAT_1740626876.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 5.

2.1.2.2 2_GPT 4.5_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run.ipynb`, `GPT_Results.ipynb`
  - CSV files (e.g., `GPT_VLAT_1740709963.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `GPT_VLAT_1740709963.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 5.

2.1.2.3 3_GEMNI 2.0 Pro_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run.ipynb`, `Gemni_Results.ipynb`
  - CSV files (e.g., `Gemini_VLAT_1741123345.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `Gemini_VLAT_1741123345.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 5.

2.1.2.4 4_GPT 4.o_Final Results with Explanation

- **Contents:**
  - Python notebooks: `VLAT_run.ipynb`, `GPT_Results.ipynb`
  - CSV files (e.g., `GPT_VLAT_1743188849.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `GPT_VLAT_1743188849.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 5, specifically for comparison with the GPT-4.o results reported by Bendeck and Stasko.

2.1.3 Additional Files

- `VLAT Questions Metadata.csv`: CSV file containing metadata about the VLAT questions
- `VLAT Questions.csv`: CSV file containing the VLAT questions and answer options

3 3_Kim_VisQA

This folder contains materials related to our Chart Question Answering (CQA) experiments based on Kim et al.'s dataset:

3.1 Data

- **Contents:** CSV files containing the data used in the CQA experiments
- **File type:** CSV
- **Software needed:** Any spreadsheet software (e.g., Microsoft Excel, Google Sheets)
- **Relation to the paper:** These files contain the dataset used for the Chart Question Answering experiments described in Section 6 of the paper.

3.2 Images

- **Contents:** PNG files of all visualizations used in the CQA experiments
- **File type:** PNG
- **Software needed:** Any standard image viewer
- **Relation to the paper:** These images are the charts used in the CQA experiments described in Section 6 of the paper.

3.3 Results

- **Contents:**
  - Python notebooks: `Kim_VisQA_results.ipynb`, `Kim_VisQA_Without Data_Claude_run.py.ipynb`
  - CSV files (e.g., `Claude_VisQA_Complete_1741226669_1.csv`) containing the model responses, time taken, and correctness
  - Corresponding DOCX files (e.g., `Claude_VisQA_Complete_1741226669_1.docx`) containing the detailed explanations from the LLM
- **File types:** IPYNB, CSV, DOCX
- **Software needed:** Jupyter Notebook, Microsoft Word or any word processor that can open DOCX files
- **Relation to the paper:** These files contain the raw data used for the analysis in Section 6.

4 Software Requirements

To fully utilize these supplemental materials, the following software is required:

1. **Python 3+** with the following packages:
   - pandas
   - numpy
   - matplotlib
   - seaborn
   - jupyter
2. **Jupyter Notebook** for running the IPYNB files
3. **Microsoft Word** or any compatible word processor for viewing the DOCX files
4. **Any standard image viewer** for viewing the PNG files
5. **Any spreadsheet software** (e.g., Microsoft Excel, Google Sheets) for viewing the CSV files

Note: The file names are self-explanatory, indicating their specific purposes within the research project. These materials allow for complete replication of the experiments and analysis described in the paper.